charged conducting spheres

This involves scenarios where electric potential applies, regarding charged conducting spheres (both hollow and solid).

Free electrons within a conducting sphere will **always** rearrange themselves to **arrive at the smallest possible energy** (within the constraints of conservation of charge); the charges **maximise their distances** from each other.

For a conducting sphere (hollow or solid) with a total charge Q:

- **outside the sphere**, E and V_e behave identically to a point object placed at the sphere's centre with a charge Q; a *tiny ass copy of the original sphere with charge Q with the position of its centre invariant would have the same E and V_e.*
- **inside the sphere**,
  - the V_e remains constant with the **potential of the surface**; $V_\mathrm{e} = \frac{kQ}{r}$, where r is the radius of the sphere.
  - the E is **zero** because there is no potential gradient inside the sphere.

Here, it is easy to recognise that V_e must be constant inside the sphere because it is a **conductor**.